

The Second Dose

A foreign protein given once may be forgiven; given twice, it meets an immune system that remembers, and the field has hardly begun to ask what follows

Almost every study of an oxygen carrier is a study of a first exposure. A patient bleeds, receives the product once or over a few days, and is followed until discharge or for a set number of weeks. The scientific and regulatory record is built on that pattern. Yet the patients who might benefit most from a carrier are not always one-time users. Some have conditions that recur: sickle cell crises, repeated bleeding from a vascular malformation, recurrent anemia in a person whose religious beliefs preclude transfusion. Some will be trauma survivors who, years later, face a second emergency. For all of them the question is what happens on the second exposure to a protein that comes from another species.

That question has received far less attention than it deserves. This essay lays out what is known, what can be reasoned, and what has to be measured, and it tries not to pretend that the answer is settled.

Why a second dose is different

The immune system meets a foreign protein in two ways. On first contact, it is a stranger, and the response takes days to develop. On the second contact, it is a familiar enemy, and the response is faster, larger and often of a different kind. This is the basis of vaccination and also of allergy.

Bovine hemoglobin is a protein from another mammal. It is similar to human hemoglobin in structure and different enough in sequence that the human immune system can, in principle, recognize it as foreign. Two things reduce the likelihood of a strong response. Chemical modification, such as the cross-linking and polymerization used in the bovine products, may alter the surface in ways that hide some sites. And mammalian hemoglobins are closely related, which may render the molecule less provocative than a protein from a distant species. Neither guarantees tolerance.

The other possibility is that the immune system does respond, producing antibodies that bind the carrier. Such antibodies could do several things. They could clear the carrier rapidly from the circulation, reducing its effect and its half-life. They could form complexes that deposit in the kidney or vessels. Or they could cause an allergic reaction of the kind that can be dangerous. Which of these happens, if any, is an empirical question.

What the veterinary experience suggests

The largest body of experience with repeated exposure comes not from human medicine but from veterinary practice, where a bovine carrier has been given to dogs and, off label, to other species for many years. The company that markets it has stated that the product has been used widely across species for over two decades. Some animals must have received it more than once, either in the same illness or in separate ones.

I do not have access to a systematic analysis of the immunological consequences in those animals, and I would be inventing if I claimed otherwise. What can be said is that the experience has not, to my knowledge, produced a widely known pattern of severe reactions on re-exposure, and that the labeling for the veterinary product has historically described it as intended for a limited number of infusions. A serious effort to answer the question would search adverse event reports, ask veterinary specialists about their experience with second doses, and, where possible, measure antibodies in animals that received repeated treatments. Such data may already exist in scattered form, and collecting it would be inexpensive relative to its value.

The human data problem

Human experience is more limited. Trials of bovine and other hemoglobin-based carriers included immunological monitoring in some cases. My understanding is that the general finding across several products was that antibody responses were infrequent and did not seem to produce clinical problems within the trials' periods. I hold that statement loosely, because I cannot reconstruct the specific studies and because the trials followed patients for short periods after a single course.

The key limitation is the design. To see a memory response, you need a second exposure after a delay long enough for immunity to develop, typically weeks to months. Most trials do not provide that. A patient who was exposed once, and was followed for thirty days, tells us nothing about what happens at day ninety. This is not a criticism of the trials, which were designed to answer other questions. It is an observation that the specific question has not been asked.

Chemistry and immunity

Some features of the products matter for immunogenicity. Polymerization increases the molecular size, and larger particles are more likely to be taken up by immune cells that present them to the rest of the system. Cross-linking with glutaraldehyde creates new chemical structures that were not in the original protein, and such structures can act as haptens, small chemical groups that provoke an immune response when attached to a carrier. PEGylation, the attachment of polyethylene glycol chains, is used in some products to shield the protein, but antibodies against polyethylene glycol itself have been described in the general literature on PEGylated drugs, and they can accelerate clearance of such drugs on repeat dosing.

The details vary by product, and the conclusion is that each product must be evaluated on its own. A finding of low immunogenicity for one chemistry does not transfer to another.

There is also the matter of purity. Preparations that contain traces of other bovine proteins, from the plasma or the red cell membrane, present a wider range of targets to the immune system. The purification steps that remove these impurities are therefore also immunological safeguards. A change in manufacturing that lets more impurity through could change the immune profile without changing anything on the label.

Endotoxin and other contaminants

Contaminants that are not proteins matter too. Bacterial endotoxin, the fragments of cell walls of gram-negative bacteria, is a potent stimulator of the immune system, and even small

amounts in a product can cause fever, low blood pressure and inflammatory signaling. Such contamination can make an otherwise well-tolerated product seem immunogenic, and it can prime the immune system to respond more vigorously on subsequent exposures. Testing for endotoxin is standard in the manufacture of injectable products, and the limits are strict. For a product made from animal blood, collected in an environment that is not sterile in the way a pharmaceutical plant is, the discipline needed is greater.

Sensitization and the transfusion analogy

Physicians who care for patients needing repeated transfusions know the problem of alloimmunization. A patient who receives many units of red cells may develop antibodies to minor antigens on donor cells, which make later transfusions more difficult to match. In certain populations, such as patients with sickle cell disease, this is a serious clinical problem that is managed by extended matching of blood.

The carrier's promise to be free of such antigens is real: without cell surface markers, there is nothing to match. But that promise does not cover the carrier's own immunogenicity. A patient who would otherwise become sensitized to red cells might instead become sensitized to a bovine protein. Whether that trade is favorable depends on the size and consequences of each. An honest description of the product would distinguish clearly between the absence of blood-group antigens, which is true, and the absence of immunogenicity, which has not been shown.

What tests would answer the question

A research program that took this issue seriously could proceed in stages, each modest and each informative.

The first is to measure. In existing samples from earlier trials, and in veterinary populations, look for antibodies against the carrier and against the chemical modification. Test their ability to bind the product and whether they change its behavior in plasma.

The second is to model. In animal studies, expose animals twice and three times, at intervals of weeks, and compare the circulating half-life and the clinical response at each dose. A shortening of the half-life on the second dose would be a signal of antibody-mediated clearance.

The third is to observe in people. In a trial with planned repeat dosing, measure antibodies before each dose and at intervals afterward. Include clinical monitoring for allergic reactions and for signs of immune complex disease, such as rash, joint pain, and changes in kidney function.

The fourth is to build a registry. For any product used under expanded access, where the physician is treating a single patient in extremity, systematic data on repeated exposure is rarely collected. A registry that logs each dose, the interval and the outcome would accumulate knowledge from cases that would otherwise be lost.

The special case of chronic use

The most challenging scenario is not the second dose but the tenth. Suppose an oxygen carrier were considered for use in a chronic condition, such as a severe anemia with no other option.

Then a patient would receive many exposures over months or years. Immune tolerance to a foreign protein can develop under some circumstances, but sensitization is equally possible, and the balance depends on dose, route and frequency. The history of protein therapeutics, including enzymes and antibody drugs produced in animal cells, is full of cases where antibodies developed in a fraction of patients and reduced efficacy over time.

The lesson is that chronic use cannot be extrapolated from acute use. It requires its own evidence, with long follow-up and immune monitoring. A responsible development plan would treat acute and chronic indications as distinct programs and would not allow success in the first to imply readiness for the second.

What the patient should know

A patient offered a carrier under expanded access should understand a few things. It is made from a foreign protein. The chance of a serious allergic reaction on first exposure is believed to be low, though not zero. The risk on later exposure is not well known. If a second dose is contemplated, the physician should be told about the first, and the patient should be observed carefully during and after the infusion, with resuscitation equipment at hand.

These points are unglamorous, and they belong in the consent conversation. The consent essays in this series argued that informed consent for an emergency product should include what is known and what is not known. Repeat dosing sits squarely in the second category, and saying so is part of respecting the patient.

Why this gap persists

One might ask why a question so basic remains open. The answer is a mixture of economics and incentives. A trial that follows patients for months after a single exposure is expensive, and a company with limited funds will choose the endpoints its regulator requires. Veterinary sales generate revenue without the obligation to collect immunological data. Academic laboratories rarely have access to the product. And the assumption that a bovine protein, chemically modified, will behave benignly has been reinforced by the absence of reports of trouble, an absence that may reflect the absence of anyone looking.

That last point is worth stressing. A silence in the record is not evidence of safety. It is evidence that no one was asked to write anything down. The remedy is deliberate: ask, record, and publish.

A modest proposal

I would propose that the community concerned with oxygen carriers agree on a minimal dataset for every exposure, whether in a trial, in expanded access, or in a veterinary clinic that is willing to share. The dataset would record the product and lot, the dose, the date, the species and the patient's identifier, any prior exposure, any reaction within a defined period, and where possible a stored serum sample. The cost is small, the value is large, and the data would answer the second-dose question within a few years of accumulating.

Such a shared record would also serve other purposes. It would support pharmacovigilance, would help detect manufacturing drift, and would give regulators a factual basis for decisions.

The Foundation could host or sponsor the coordination, since the work is neutral, unglamorous and beneficial to every party.

Reading an antibody result

Suppose a study does measure antibodies. What would the result mean? A positive test means that the patient's serum contains something that binds the carrier. It does not by itself mean harm. Many people carry antibodies to many foreign proteins without symptoms, and binding in a laboratory dish is not the same as clearance in the body. The clinically relevant questions are graded. Do the antibodies neutralize the function of the carrier? Do they speed its removal? Do they form complexes that lodge in tissue? Do they accompany symptoms?

A careful analysis therefore pairs the antibody test with a functional one, such as the circulating half-life after a repeat dose, and with a clinical one, such as the incidence of allergic reactions. A result that shows antibodies with no functional or clinical consequence is reassuring. One that shows a shortened half-life without symptoms is a practical problem, since the product would work less well the second time. One that shows symptoms is a safety signal. Reporting all three together prevents both false alarm and false comfort.

Timing and the window of vulnerability

Immune responses follow a pattern in time. Antibodies of the first kind, which appear within about a week or two of exposure, are followed by a more mature response over the next weeks. If a second dose arrives during the transition, when antibody levels are rising, immune complexes are most likely to form. If it arrives long after, when antibody levels have fallen but memory cells remain, the response can be swift and strong. The riskiest interval may therefore not be the shortest or the longest but somewhere between, and it is unlikely to be the same for every patient.

For a clinician, the practical consequence is that the date of the first exposure is a clinical fact that belongs in the record, easily accessible when a second is considered. Emergency care rarely provides for this. A patient treated in one hospital may arrive months later at another, with no link between the records. The registry proposed above would help, as would something as simple as a wallet card given to every recipient stating the product, the date and the lot.

Species and the choice of a source

The question of immunogenicity feeds back into the choice of raw material. Bovine hemoglobin is one option. Others include hemoglobin from pigs, from horses, from human donors and from engineered microbes producing human sequences. In principle, a molecule with the human sequence should be less immunogenic than one from a cow. In practice, recombinant proteins produced in microbes can differ from natural ones in modification and folding, and chemical stabilization changes the surface regardless of the source.

One reasonable view is that the immune question is a reason to keep several sources under study rather than to commit early to one. If repeat dosing turns out to be a problem for bovine products, a human-sequence alternative may be the only route for chronic use. If it turns out not to be, the bovine product retains its advantages of scale and experience. The information will only arrive if someone looks, and until it does, diversification is prudent.

What regulators could reasonably ask

A regulator reviewing an oxygen carrier for approval could ask for immunogenicity data proportional to the intended use. For a single-dose emergency indication, the demand for repeat-dose data might reasonably be modest, with a commitment to post-approval surveillance. For an indication that implies repeated exposure, the standard should be higher, and evidence from repeat-dose studies should be required before approval. Splitting the requirement by indication would avoid both extremes: burying an emergency product in paperwork and approving a chronic one on evidence collected for a different purpose.

Post-approval surveillance deserves a special mention. Products approved under conditions of unmet need, with a commitment to further study, have a mixed record on fulfilling the commitment. A public schedule of the promised studies, with the results posted when they are due, would make the arrangement more than a formality.

Closing thought

The first dose of a carrier is a bet on the moment. The second is a bet on memory, the body's and ours. The moment has been studied for a century, and memory has hardly been studied at all. If the field is to mature from emergency improvisation to reliable medicine, it must learn what its products do to the immune system that meets them twice. The experiments are not exotic, and the money required is small compared with a pivotal trial. What is needed is the decision to look.

Further reading

The immunogenicity of protein therapeutics and of PEGylated drugs is reviewed in the general pharmaceutical literature, including work on anti-polyethylene glycol antibodies. Immunological monitoring in hemoglobin-based oxygen carrier trials is described in individual trial reports and in reviews of the class published in transfusion and artificial cells journals. Alloimmunization in sickle cell disease is reviewed in hematology literature. The veterinary product's labeling and the company's statements on its use across species are available from the manufacturer. Claims above about the absence of reported problems reflect my limited reading and should be checked against adverse event databases.